

Simulation-in-the-Loop Thermal-Aware Task Remapping for Silicon-Photonic Networks-on-Chip

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Abstract—Thermal management is a key barrier to system-level silicon-photonic networks-on-chip (ONoCs). In task-level workloads, task-to-processing-element (PE) mapping shapes heat-source placement, DVFS throttling, optical paths, wavelength activity, SOA activation, and microring-resonator (MRR) thermal-tuning demand. However, existing ONoC thermal-management methods mainly optimize device tuning, wavelength assignment, routing, or topology control, leaving task mapping underused as a system-level control variable. This paper therefore treats ONoC thermal management as a joint tradeoff rather than only chip-temperature reduction or MRR drift compensation. We propose a simulator-in-the-loop thermal-aware task remapping method based on a genetic algorithm. Each candidate mapping is evaluated by a full-system OMNeT++ model with an 8-wavelength WDM optical layer, dynamic MRR tuning, SOA and laser energy, a compact RC thermal network, and DVFS feedback. The optimizer minimizes a baseline-normalized cost combining peak temperature, temperature imbalance, hot-PE count, makespan, communication cost, congestion, DVFS penalty, load imbalance, and total energy. Across GEMM, MPEG4, VOPD, and HNN DAG workloads, the method reduces the composite cost in all cases. It improves thermal, performance, communication, and energy metrics for GEMM and MPEG4; reduces VOPD makespan and energy by 51.6% and 33.0%; and cuts HNN hot PEs from 16 to 4. The results show that task remapping is a practical system-level thermal-management mechanism for ONoCs when thermal stability, photonic compensation, performance, and energy are optimized jointly.

Index Terms—silicon photonics, optical network-on-chip, thermal-aware task mapping, microring resonator, genetic algorithm, OMNeT++

I. INTRODUCTION

Silicon-photonic networks-on-chip (ONoCs) are a promising interconnect substrate for many-core processors because they provide high bandwidth density, wavelength-division multiplexing (WDM), and low-latency data movement [1], [2]. At the system level, ONoC execution is determined by both task execution on processing elements (PEs) and communication among dependent tasks. These two factors jointly shape runtime performance, thermal behavior, and optical-layer energy.

Task-to-PE mapping determines where computation is placed on the chip and therefore defines the spatial distribution of heat sources. This heat distribution affects PE hotspots, DVFS throttling, and task execution time. It also affects the temperatures seen by nearby microring resonators (MRRs), whose resonance wavelengths are sensitive to temperature variation [3]. Temperature-induced shifts can move MRRs away from their target WDM channels, increasing the need for dynamic thermal tuning to maintain MRR alignment. Thus, thermal behavior in an ONoC has both electronic and photonic consequences.

Task communication introduces another coupled effect. A mapping changes which task pairs communicate across the chip, which physical paths are used, and how WDM optical transmission activity is distributed over time. It therefore changes wavelength-channel activity, SOA activation time, and optical-layer energy. A mapping that reduces peak temperature may increase communication distance or optical activity, while a mapping that improves locality may concentrate heat and increase photonic-device compensation cost.

Consequently, ONoC thermal management is not only a matter of compensating MRR thermal drift, and it is not only a matter of lowering chip temperature. It is a system-level multi-objective tradeoff among task mapping, heat distribution, MRR alignment, WDM optical transmission activity, performance, and energy. This view is important because optimizing any single metric can hide adverse effects on another part of the thermal-optical-performance chain.

Existing ONoC thermal-management techniques mainly focus on device tuning, wavelength assignment, routing, or topology control [4]–[6]. These techniques are necessary, but they often treat task-to-PE mapping as a fixed input. This leaves an important system-level variable underused: by changing where computation and communication originate, task mapping can reshape both the thermal field and the optical activity pattern before lower-level compensation mechanisms are applied.

This paper proposes a simulator-in-the-loop thermal-aware task remapping framework for ONoCs. The framework uses a genetic algorithm to search task-to-PE mappings. Each candidate mapping is evaluated by a full-system OMNeT++ model that includes task execution, PE power, a compact RC thermal network, DVFS feedback, WDM optical transmission activity, MRR dynamic thermal tuning, SOA pump power, and laser energy. The resulting thermal, optical, performance, and energy metrics are combined through a baseline-normalized multi-objective fitness function.

The main contributions of this work are as follows. First, we formulate task-to-PE mapping as a system-level thermal-management variable for ONoCs, linking heat-source placement with PE throttling, MRR alignment, photonic-device compensation cost, and optical-layer energy. Second, we introduce a simulator-in-the-loop search framework that evaluates each candidate mapping using the same full-system OMNeT++ model used for validation. Third, we evaluate the method on four DAG workloads and show workload-dependent tradeoffs among thermal behavior, WDM optical transmission activity, performance, and energy.

II. METHOD

Figure 1 placeholder: insert the method framework figure here. The final figure should show the closed-loop flow from DAG workload and initial/reference mapping, through GA candidate remapping and full-system OMNeT++ evaluation, to normalized objective terms and next-generation selection. To be written: open the method section by using Figure 1 as the in-text roadmap of the proposed simulator-in-the-loop task remapping framework. The text should explain the data flow, feedback loop, and why full-system simulation is used inside the search.

A. System Model

To be written: define the simulated ONoC platform, including the 4×4 PE array, 8-wavelength WDM optical layer, MRR tuning, SOA/laser energy accounting, compact RC thermal network, and DVFS feedback. This subsection should make clear which physical and system-level effects are evaluated in OMNeT++ and which assumptions bound the model.

B. Task Remapping

To be written: formulate the DAG workload, task-to-PE mapping vector, initial/reference mapping, and GA individual representation. This subsection should explain how a candidate mapping is converted into a temporary workload configuration and evaluated by the same simulator used for final validation.

C. Multi-Objective Fitness

To be written: introduce the initial-mapping-normalized composite objective. The text should define the nine terms: peak temperature $T_{\max}$, temperature imbalance σ_T , hot-PE count N_{hot} , makespan, communication cost, congestion proxy, DVFS penalty, load imbalance, and total energy. It should also state that the weights express the paper's full-system design priority rather than being tuned post hoc to favor one method.

III. RESULTS

To be written: open the section with the experimental setting, workload list, and comparator definitions. Use "ReferenceMapping" only as a normalization anchor / initial mapping, while treating Thermal-SA-TAS and CommAware-Heuristic as external baseline methods with narrower native objectives.

Figure 2 placeholder: insert the full-objective comparable score figure here. It should be discussed as a comparable score under the proposed full-system objective; ReferenceMapping is the normalization anchor, not a baseline method.

To be written: use Figure 2 to report the overall score ranking. The paragraph should emphasize that Full-GA directly optimizes this objective, so the figure demonstrates its behavior under the proposed design objective but should not be used alone as a universal fairness claim.

Figure 3 placeholder: insert the workload-level normalized-term decomposition here, with panels for GEMM, MPEG4, VOPD, and HNN. The figure should break the objective into thermal safety ($T_{\max}$, σ_T , and N_{hot}), performance (makespan

and DVFS), communication pressure (communication cost and congestion proxy), mapping balance (load imbalance), and energy (total energy).

To be written: use Figure 3 to explain where the score reduction comes from. GEMM and MPEG4 should be framed as favorable cases with broad improvements. VOPD should be framed as a performance/energy-oriented trade-off, without claiming that all thermal terms improve. HNN should be framed as the main multi-objective tradeoff case: hotspot/DVFS-related safety improves, while makespan can worsen relative to the initial mapping.

IV. CONCLUSION

To be written: close with four points: the contribution of simulator-in-the-loop task remapping for ONoC thermal management, the decisive evidence from the full-objective score and metric-level decomposition, the implication that mapping can act as a system-level control variable alongside device-level tuning/routing/wavelength assignment, and the boundary that gains are workload dependent and should not be stated as uniform improvement across all metrics.

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